

MONTHLY PROGRESS REVIEW

AUGUST 2026

Project ID: MKCE/R&I/Consultancy/2026/001

AI-Based Egg Tray Inventory Management System (EG-Counter)

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1. Monthly Progress Summary

A working Android and Cloudflare/Roboflow counting route was established and measured against ten visually reviewed warehouse photos. The evaluation exposed large view-dependent errors; the app withheld a final count for an inconsistent three-view scene.

Current phase: model evaluation and mobile/backend integration. The 31 August measurements are a per-photo detection baseline, not certified egg-filled inventory.

2. Work Completed This Month

2.1 Prototype and inference integration

The Android client sent photographs to a Cloudflare Worker, which called the deployed Roboflow RF-DETR Medium model projec-mutta/2. Direct Roboflow and Worker counts matched on all ten evaluated frames. The client parsed the backend response without changing the tray count.

2.2 Ground-reference and evaluation work

Ten images received manual visible-layer references; four uncertain photos were excluded. The frozen baseline was 2/10 exact images (20.0%), 22.9 trays mean absolute error, and 58.43% mean per-image count agreement. Count agreement means 100 times (1 minus absolute count error divided by the reference); it is not detection precision or warehouse accuracy.

2.3 Photo evidence and measured counts

Saved photo / case	Manual visual reference	Model / mobile pathway	Count agreement
img05, six-stack wall	120	124	96.67%
img04, brown two-stack scene	32	29	90.62%
img07, single evaluated frame	21	21	100.00%
All 10 accepted frames	10 reviewed references	2 exact; MAE 22.9	58.43% mean

The numbers above are per-photo backend counts in the mobile app path. They are not accepted three-view inventory totals. The retained photos may have been captured before August; this table dates their evaluation, not capture.



Original evaluated photo: 120 manually visible tray layers.



Saved V2 detection overlay: 124 per-photo detections.

3. Challenges Faced

- One 60-layer scene yielded 9, 12 and 93 across three views; the app withheld a final total and requested a rescan.
- Dense stacks caused duplicate boxes; distant or dim stacks caused misses.
- Visual layers do not certify concealed egg occupancy. The benchmark had no independent physical recount.

4. Plan for September

- Audit annotations and scene overlap across data splits.
- Compare angle, crop and threshold methods against per-stack references.
- Test quality prompts and multi-view counting on untouched, physically counted scenes.

5. Overall Project Status

Phase	Status at 31 August	Measured progress
Dataset and visual references	Baseline established	10 accepted / 4 excluded
Model and backend	Integrated; inaccurate on varied views	2/10 exact; MAE 22.9
Android integration	Prototype operational	Response parsing verified
Exact warehouse inventory	Not verified	No physical acceptance set

5.1 Overall Progress

Prototype integration and baseline evaluation were complete. Overall project completion cannot be quantified. The next milestone is an independent physical scene recount with exact stack counts and rejection coverage.

Evidence: accuracy-evaluation/report.md, summary.json, results.csv, ground-truth-exclusions.csv, and saved img05 original/overlay.